Content recommendation based on recognised Emotion

Sireesha Challa
Department of Computer Science and Engineering
Sathyabama Institute of Science and Technology
Chennai,India
sirichalla29@gmail.com

Gowri Manohari V.
Department of Computer Science and Engineering
Sathyabama Institute of Science and Technology
Chennai,India
gowrimanohari.cse@sathyabama.ac.in

Roobini M.S
Department of Computer Science and Engineering
Sathyabama Institute of Science and Technology
Chennai,India
roobini.cse@sathyabama.ac.in

Nandhini P.
Department of Computer Science and Engineering
Sathyabama Institute of Science and Technology
Chennai,India
nandhini.p.cse@sathyabama.ac.in

Daphine Desona Clemency C.A
Department of Computer Science and Engineering
Sathyabama Institute of Science and Technology
Chennai,India
daphine.cse@sathyabama.ac.in

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***Abstract*— E-commerce, video websites, and social networking sites are just a few examples of where recommendation algorithms are currently being used. As the amount and variety of data used in recommendation systems grows, more and more of it will be unstructured multimodal data, including text, images, and videos. Yet, it is difficult to employ unstructured multimodal data to enhance the effectiveness of recommendation schemes because of the illustration gap between different modalities. In this paper, we present a new multimodal framework for user-driven ensemble modelling based on the recommendation of mood descriptions. In contrast to standard approaches, our video reference formulation makes use of not just the audio stream but also the face (emotion) and textual modalities that together make up a video document's elementary data. Audio data is used as input by algorithms like Gaussian Naïve Bayes (GNB), Logistic Regression (LR), and Convolutional Neural Network (CNN), whereas text data is used by methods like, LR, Long-short Term Memory (LSTM), and Naive Bayes. The user's current disposition is used to inform recommendations for appropriate music, quotes, etc. generated by these ensemble models. Comprehensive experiments on real-world datasets demonstrate that, across a wide range of assessment measures, the proposed ensemble model greatly outperforms the state-of-the-art baselines.**

Keywords*— *Recommendation systems; Long-short Term; Neural Network; Unstructured multimodal data; User-based ensemble model.

I. INTRODUCTION (*HEADING 1*)

As the Internet expands and processing power improves, more and more items and contents are being made available online [1]– [3]. Furthermore, consumers now have access to data in a wide variety of formats, including text, picture, and video. The problem of information overload [4] is exacerbated by the glut of available goods, which includes clothing, media, and literature. The development of search engines and recommendation systems is a direct response to this issue. In terms of providing consumers with tailored experiences and generating revenue, recommendation systems are far more engaged than search engines. Users' growing concern for privacy [5] means that most websites cannot collect specific user traits; yet, there is more data available regarding content and items, making a hybrid recommendation system the superior option. In most online systems, each user is linked to a collection of consumables that can be monitored over time [6].

Research into emotional computing in general, and feeling acknowledgement in particular, is becoming a burgeoning and promising study area [7] as scientists work towards the eventual aim of imbuing machines with feelings. Many different types of technology, including e-health, systems, have recently made use of emotion-aware intelligent systems (e.g., chatbot). Some types of intelligent systems could benefit greatly from the incorporation of computer-based automatic emotion identification, such as those used for online gaming, neuro-marketing (the evaluation of consumer feedback), and mental health monitoring. Yet, the emotional qualities of media material have not received the attention they deserve, nor have they been fully used in real-world or commercial contexts, despite being an integral part of this data [8]. Humans can't function well without the ability to express and comprehend the emotions of others. Furthermore, the usefulness and flexibility of multimedia content and multimedia investigation could be improved by including emotional information.

Despite some progress in this area (such as emotion examination of users' audio or video for enriching users' profiles for activity recommendation, affect prediction from movies, or speech [9]), it has not been implemented in practise because reproducing such algorithms is labour-intensive and error prone. Research hotspots for Collaborative Filtering (CF) are primarily focused on neural CF approaches [10-12], thanks to the enormous success of deep learning in Computer Vision (CV) and (NLP). When compared to older versions of CF like neighbour-based CF and Matrix Factorization, the state-of-the-art approaches that make use of deep neural networks have shown significant improvement (MF). Deep learning techniques for CF include modelling recommendation algorithms and learning feature representations. Learning accurate representations of items, either in single- or multi-modal formats, is crucial for recommendation [13]. The common underlying information between various modalities has thus far been disregarded by most existing approaches, which instead use individual neural network models to learn the properties of each

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modality and then create a common spatial representation by simple weighted totalling.

Ensemble machine learning and deep learning models are taken into account for advising in this study. The user's mood is taken into account before advising an activity by analysing data from the user's face, the user's text, and the user's audio. The layout of the following pages is as follows: In Section 2, we describe the relevant literature, and in Section 3, we explain the proposed model. Sections 4 and 5 contemporary the results of the experiments and their analysis.

II. Related works

Using user-post interaction evaluations, implicit response, and supplementary information, Jiang et al. [14] offer a hybrid recommendation model based on bottomless emotion analysis and multi-source perspective fusion to generate individualised recommendations. The data set is processed with the HITS algorithm, which can identify highly influential users and messages while simultaneously removing the vast majority of irrelevant ones. Second, we address the challenge of making use of the recommendation system's text description information of post content by designing a calculation method for quantifying the similarity of candidate posts and a method for determining K nearest neighbours. In order to reduce the data distribution deviation introduced to the training data pool during iterative training, we then apply the cooperative training technique to achieve the fusion of two proposed viewpoints. Lastly, the research concludes with a comparison of the DMHR algorithm's performance on Twitter with that of other state-of-the-art algorithms. The trial findings reveal that the DMHR algorithm significantly enhances the performance of score prediction and recommendation.

A dynamic music selection system based on human emotions has been developed as a prototype by Shalini et al. [15]. Songs are trained for each emotion based on how each individual listens to music. Using learning methods, this system can determine a user's mood based on an input image, and then play them a song that fits their current state of mind. By appealing to users' emotions, this method brings a more human dimension to the software. As a result, our planned system uses computer vision and machine learning to learn how to identify human emotions for the purpose of creating an emotion-based music player. We utilise openCV to recognise emotions and recommend songs in our experiments.

An classification framework (EMRCF) has been proposed by Quasim [16], which is able to accurately classify songs based on an individual's interpersonal team's recollection and emotional state. Specifically, technologies must be created that instantly classify the characters based on people's feelings while adding new tracks to an IoT app application. Among the most important questions in project management is this one. Emotional data can be uncovered through study using the empathetic framework. Task force conversations in a micro-enterprise can have musical qualities. Dynamic designation is accomplished with the use of a neural network and correlation analysis. The proposed revolutionary prediction accuracy effectively categorises songs and recognises the majority of emotional responses triggered by music audiences. Decision trees, deep cognitive systems, neighbour-closest, and relevance vector machines are some of the proposed algorithms compared in this research. The EMRCF achieves a level of prediction accuracy of 96.12% and a precision rate of 96.69% that is unmatched by alternative methods.

A more enhanced Deep Residual Network (ResNet-38) for user emotion identification has been introduced by Zhang [17], with an accuracy of 64.02 percent on the Kaggle-fer2013 testing set. Classifier SVM and four-layer convolutional neural networks (CNN) are two more common approaches that yield average accuracies of 40.7% and 60.7%, correspondingly. So, it is reasonable to state that our model is superior than more conventional alternatives. standard fare of sentimental media and music Web scraping of the user's personal information, such as his or her recent remarks, is used in recommendation systems, and from this data, researchers build "interest models" to deduce the user's state of mind. Though thorough, this approach invades users' privacy, yields skewed data, and is slow and unresponsive. To address these shortcomings, the purpose of this study is to present a novel approach to the Recommendation system.

To address users' emotional requirements and boost their mental health, Liu et al. [18] created a novel emotion-based personalised music recommendation outline. We built an LSTM-based model into our framework that can predict a user's emotional state and play them music that best fits their present state of mind. To ensure that the recommendation actually helped users' mental health, a care factor was applied to the results. Our innovative system provides accurate and helpful recommendations, as demonstrated by our experiments and user study.

Using a user's feelings and preferences, Jaiswal et al. [19] set out to create an effective recommendation system. None of the aforementioned information is necessary for the suggested system to function, and neither is the user's constant and undivided focus. In this framework, we use low-cost, visible-light "webcams" to record users' eye-gaze and facial expressions while they navigate the web. Pupil-center extraction of both eyes is used in the eye-gaze detection method, with the reference point determined by a combined probability calculation. Facial landmark points are used to deduce the user's state of mind. Both techniques operate in near real-time, allowing the projected system to give timely, insightful recommendations independent of user comments and purchasing histories.

Bandyopadhyay et al. [20] have built an effective recommendation system that takes into account the user's facial expressions. A camera records a video of the user's expression. Users' emotions can be analysed with the use of facial expression recognition. Without using ratings or past purchases as anchors, the proposed approach creates a dynamic, intelligent recommendation system. This method is employed to anticipate a consumer's reaction prior to product selection or purchase by analysing facial traits associated with a given emotion. Although a customer's true feelings about a product might not be expressed when they're rating it, the results of the experiments suggest that the proposed method can result in a superior online recommendation system by capturing the reaction in real time. Based on how it performs in the actual world, the experimental findings show that it is reliable.

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III. Proposed System

The model 's workflow is divided into three modules to find the user's emotion. The results are passed to the prediction module, where the content songs and quotes are suggested to the user depending on the detected mood. Figure 1 presents the workflow architecture.

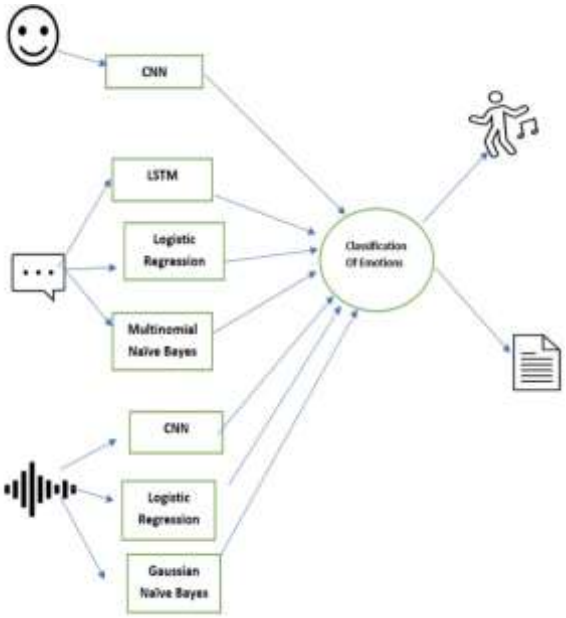

Figure. 1. Proposed System's Flow.

1) Emotion detection from face

Reading Dataset: - The dataset used in this module contains images of human facial expressions. The total number of images are 29689, these images are classified as angry, disgust, fear, happy, sad and surprise according to the facial expressions.

2) Convolutional Neural Network Modelling

This is a subclass of deep neural networks. The input image goes through many layers in this modelling for feature extraction. Firstly, the input images are reshaped if necessary then the weightage is adjusted followed by adding layers of zeros to the input image this process is called padding. Rectified linear activation function is used in this model this layer checks whether the neuron should be activated or not. The final output of this modelling is the classification of the images into seven moods namely angry, disgust, fear, happy, sad and surprise.

3) Procedure for Training and Testing

The data set has been divided into training data consisting of 28821 images and testing data with 7066 images. Major portion of the data is used for training purposes. Batch normalization technique is used to increase the learning rate. The addition of dropout layer reduces the overfitting.

4) Evaluation:

The correctness of the module is validated using the classification report and visualized using the confusion matrix.

B. Emotion detection from text

In this module to detect the user's mood from the text three algorithms have been taken into account.

Reading Dataset: - The text needed for training purpose is taken from twitter data. The data set consists of 40,000 tweets with its identity, content and the sentiment associated with it.

Data Pre-processing: - The content taken from twitter has to be transformed to make the task easier. Generally, in twitter the enjoyer uses both text as well as emojis, therefore for the analysis the emojis will be converted .Along with it website links, symbols will also be replaced. Using Natural Language Processing Toolkit, meaning full words from the content is taken out using the process of tokenization and lemmatization. The outcome is encoded to be machine readable.

1) Ensemble Classifiers

a) Naive Bayes algorithm

Naive Bayes (NB) is a classification algorithm-based probability and statistical technique. With its ease of use in making all features equally weighted in the final conclusion, this method has become a mainstay in machine learning applications. This ease of use is matched by the NB method's computational efficiency, making it a promising tool with broad applicability. The prior, posterior, and class conditional probabilities are the foundation of the NB classification. Among them are its simplicity for large datasets and its practicality. It could be applied to categorization issues with both two and more classes. Discrete and continuous data can be trained with the same minimal amount of data. This technique has potential use in spam filtering and document classification.

b) Logistic Regression

In order to solve a binary classification problem, the supervised learning algorithm logistic regression is employed. Many more involved extensions are available for the logistic regression model, which is a mathematical framework for modelling binary classification using the logistic function. Logistic regression is a type of regression model that is used to make predictions about the class to which a particular data point or entry most likely belongs. In order to model the data, logistic regression employs a sigmoid function. Logistic regression's many strengths include its ease of implementation, efficiency both computationally and in terms of training, and its amenability to regularization. No scaling is required for input features. Yet, nonlinear problem-solving abilities and susceptibility to overfitting.

c) CNN Model

We recommend utilizing CNN-based models to classify user emotions. The most salient features of the output are captured by convolution and max pooling.

Pooling layers, Convolution layers, and fully connected layers are the three basic building blocks of a (CNN). A completely linked layer incorporates the features extracted by preceding layers (such as Pooling and Convolution) into the final result (such as classification).

The primary layer of a convolutional network is the convolutional layer. The input is slid across the filter to extract features. The goal is to identify a predefined set of characteristics within the incoming images.

Pooling Layer: This layer is used to reduce the size of the feature maps. This reduces the amount of work required to calculate the network and the sum of parameters that need to be learned.

At the very end of a convolutional neural network (CNN), there is a layer recognized as the Fully Connected Layer,

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which receives its input from the output of the final Pooling Layer. Fully linked layers use the features acquired from earlier layers to perform the classification.

d) Long Short-Term Memory

Many researchers have successfully implemented their own methods for identifying users' emotions expressed in online discussion boards. We require a large amount of labelled data that has been segmented into different groups in order to detect emotions. The standard method involves preprocessing the dataset and employing multiple machine learning models. Input gate, output gate, and forgets gate are the three gates that make up an LSTM. Input gates are responsible for incorporating new data into the current state of a cell, whereas forget gates decide which bits of data are necessary and which can be forgotten. Based on the state of the internal cell, the output gate decides what output to produce.

IV. Results and Discussion

A. Evaluation Metrics

Our method's efficacy and classification accuracy can be evaluated with the use of assessment metrics. Evaluation criteria include precision (P), and accuracy (A) (AC). Differentiating factors in the arrangement, like:

$$P = \frac{tp}{tp+fP} \quad (1)$$

$$SE = \frac{tp}{tp+fn} \quad (2)$$

$$SP = \frac{tn}{tn+fp} \quad (3)$$

$$\mathrm{AC} = \frac{tp+tn}{tp+fp+tn+fn} \quad (4)$$

Where tp, tn, fp, and fn denote the summation of cases such as genuinely positive and negative, false positive and negative.

B. Confusion Matrix Representation for ensemble model

Figure 2 presents the confusion matrix for audio model, where a) represents the analysis for LR and b) represents the analysis of NB.

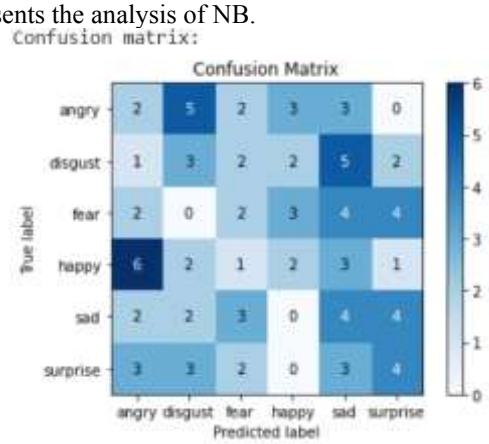

Figure 2: (a). AUDIO_LR.

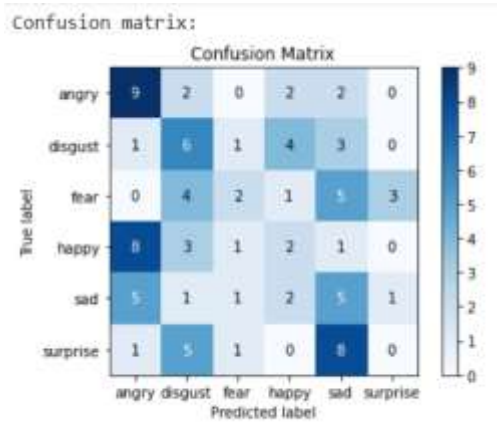

Figure 2: (b). AUDIO_NB

Figure 3 provides the confusion matrix for text data, where a) signifies the confusion matrix for NB and b) represents the confusion matrix for LR. Figure 4 presents the confusion matrix for LSTM model.

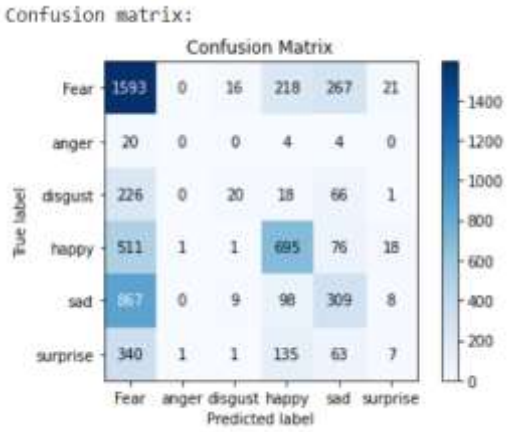

Figure 3: a) TEXT_NB.

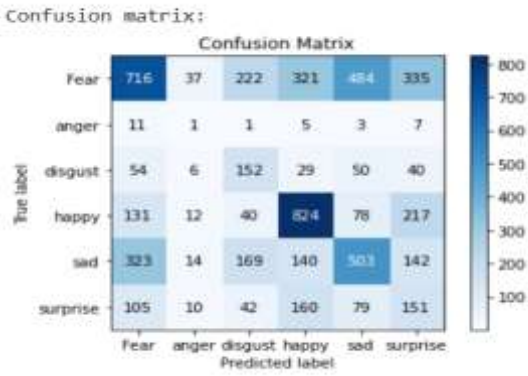

Figure 3: (b). TEXT_LR.

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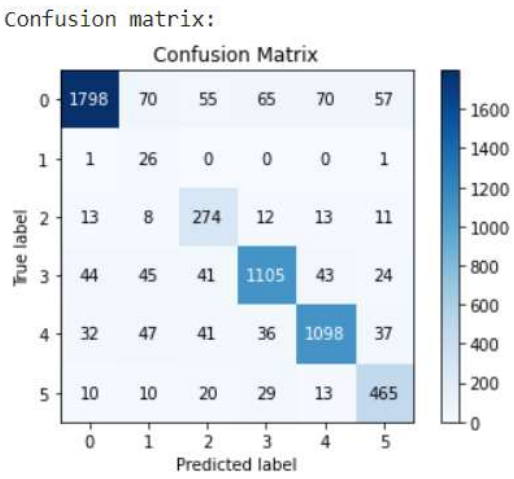

Figure 4: Text_LSTM

Table 1 presents the screenshot of our implementation work, which is taken for three different modalities.

Table 1: Relative investigation of Projected Model

Methodology	Parameter Evaluation			
	Accuracy (%)	Precision (%)	Recall (%)	F-measure (%)
NB	88.89	79.12	80.92	85.27
DT	72.32	80.53	83.69	86.07
LR	81.43	82.07	90.06	89.28
CNN	87.16	81.04	84.17	83.08
LSTM	94.38	95.43	96.46	96.34
Proposed Ensemble	**96.90**	**97.84**	**98.20**	**98.67**

In the investigation of accuracy, the NB achieved 88%, DT achieved 72%, LR attained 81.43%, CNN has 87.16%, LSTM achieved 94.38% and proposed model achieved 96.90%. When comparing with all models, NB achieved low performance, i.e., 79% of precision, 80.92% of recall and 85.27% of F-measure. The other single models such as DT, LR and CNN achieved nearly 83% to 89% of precision, recall and F1-score. But the LSTM model achieved 95% of precision, 96% of recall and F1-score, where the projected ensemble model achieved 97.84% of precision, 98.20% of recall and 98.67% of F1-score. Figure 5 and 6 presents the graphical analysis of proposed model.

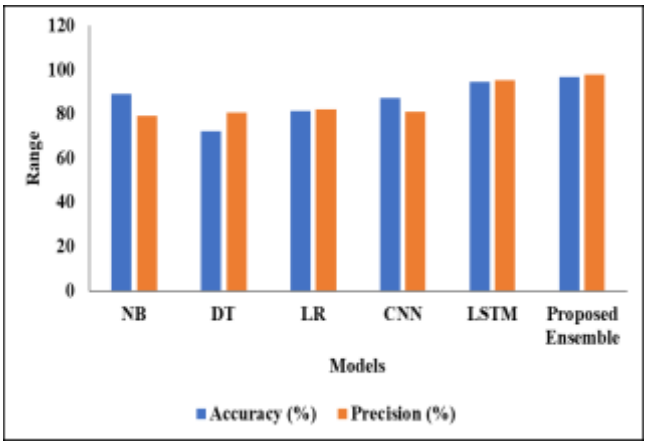

Figure 5: Graphical Analysis of Proposed Model

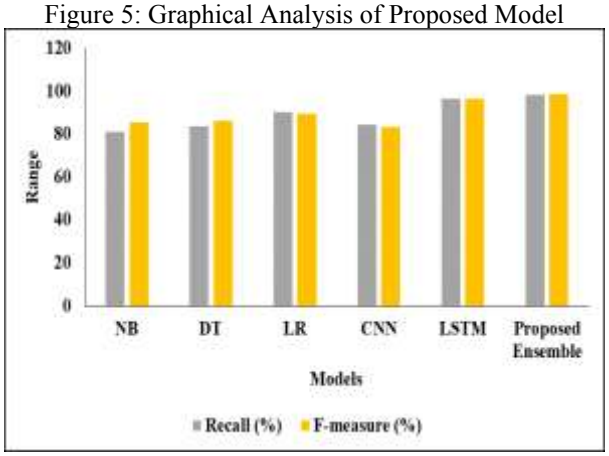

Figure 6: Performance of Proposed Ensemble Model

V. Conclusion

Here, an ensemble model is created to predict a user's emotions and then prescribe an appropriate course of action. Audio, text, and facial expression data are all utilised in this procedure. The user is then given suggestions for additional songs and quotes to complete based on the emotion. According to the results of the experiments, the ensemble model outperforms the single model on a number of different measures, including accuracy, precision, recall, and F1-score. The research makes use of both Twitter data and facial expressions. The study paves the way for using current characterization data to create high-performing computer models, which can then be used to speed up the formulation development process. Emotion impacts nearly every aspect of human communication, including the auditory (facial expression, voice, gesture, posture, etc.), kinaesthetic (EEG, RSP, etc.), and visual (goal, preference, environment, social setting, etc.) modalities. Future research needs to address the difficulty of studying multi-modal emotion identification. Also, only a small amount of data is used for validation, therefore future work will need to broaden the scope of available data to include social media sites like Facebook, YouTube, etc.

The dataset used in work is taken from the publicly available.

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